

REVIEW

Management of Diuretic Induced Hypokalemia in Elderly Hypertensives

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ABSTRACT: Hypokalemia occurs frequently in elderly hypertensives treated with diuretics, where it enhances cardiovascular, cerebrovascular, and renal risk. Literature Review. We review the evidence that hypertensive patients with diuretic-induced hypokalemia are at special risk from increased penetration of aldosterone into the brain, enhanced mineralocorticosteroid receptor and postreceptor signaling, and enhanced activation of the mineralocorticosteroid receptor by cortisol. This leads to unique considerations for the preferred strategy to prevent or treat hypokalemia in this vulnerable group.

Key Words: aged ■ brain ■ diuretics ■ humans ■ hypokalemia

Renal potassium (K^+) excretion is increased by loop, thiazide, and thiazide-like diuretics because of an increase in K^+ secretion from increased tubular fluid flow and sodium delivery to the distal nephron, leading to a fall in serum K^+ concentration (SK) and an increase in the frequency of hypokalemia. In addition, diuretics increase plasma aldosterone concentration, which activates the mineralocorticosteroid receptor (MR) in the distal nephron to enhance K^+ excretion. Hypokalemia is usually defined as SK <3.5 mmol/L. However, patients with a milder degree of hypokalemia of 3.5 to 4.0 mmol/L also have a significantly increased risk of short-term mortality.¹ This broader definition of hypokalemia increases substantially the number of elderly hypertensive patients at risk because of suboptimal levels of SK. The increase in plasma aldosterone concentration and the metabolic alkalosis provoked by diuretics distribute K^+ into cells. Thus, diuretics can cause hypokalemia without decreasing total body K^+ .²

INCIDENCE AND CONSEQUENCES OF HYPOKALEMIA WITH DIURETICS

As reviewed recently, diuretic-induced hypokalemia is frequent and associated with many adverse outcomes in patients with heart failure and chronic kidney disease (CKD)² or hypertension.³ The Systolic Hypertension in

the Elderly Program reported a dose-dependent fall in SK of 0.5 mmol/L in elderly hypertensive randomized to chlorthalidone. Among those in the active treatment group, a post hoc analysis revealed that the risk of cardiovascular events was about 50% greater among those who experienced hypokalemia.⁴ The risks associated with hyperkalemia (SK >5.0 or 5.5 mmol/L) in patients with CKD and diabetes are well established. However, hypokalemia is more prevalent than hyperkalemia, even in CKD, and is associated with a similar or greater risk of cardiovascular events, yet is seldom treated or prevented.² While replacement of K^+ is recommended for treatment of hypokalemia, this is often not successful in restoring a normal level of SK.⁵ Thus, new therapeutic approaches are required. We review the evidence that preventing or treating hypokalemia with a mineralocorticoid receptor antagonist (MRA), either alone or in combination with K^+ replacement, would be beneficial in elderly or very old (>80 years) hypertensives treated with diuretics to reduce adverse cardiovascular, renal, and cerebrovascular outcomes.

RATIONALE FOR THE USE OF AN MRA

A study of 5100 hypertensive patients referred to a tertiary hypertension unit found that 28% of those with hypokalemia had primary aldosteronism (PA). However,

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Nonstandard Abbreviations and Acronyms

11-HSD2	11-beta hydroxysteroid dehydrogenase type 2
BBB	blood-brain barrier
CKD	chronic kidney disease
CNS	central nervous system
K⁺	potassium
MR	mineralocorticosteroid receptor
MRA	mineralocorticoid receptor antagonist
PA	primary aldosteronism
SK	serum K ⁺ concentration

44% of the patients with hypokalemia in this study were receiving diuretic therapy, which complicated the interpretation of these findings. In this large study, hypokalemia was associated with cardiovascular disease events, CKD, and metabolic syndrome independent of PA, while PA was associated with cardiovascular events, CKD, diabetes, metabolic syndrome, left ventricular hypertrophy and microalbuminuria independent of hypokalemia. The authors concluded that hypokalemia and PA are independently associated with the occurrence of cardiovascular events.⁶

There are special problems in diagnosing PA in the elderly since aging decreases plasma aldosterone concentration⁷ and there is an increase in the prevalence of incidental adrenal adenomas, adrenal aldosterone-producing cell clusters, and low-renin autonomous aldosterone production⁸ even when plasma aldosterone concentration levels are not increased.⁹ Moreover, the expression of the MR in human blood vessels is increased by reactive oxygen species, which increases with age.¹⁰ Increased MR signaling in vascular smooth muscle cells underlies age-dependent increases in blood pressure in mice and drives angiotensin-induced vascular oxidative stress.¹¹ Activation of the vascular MR, as during visceral obesity, is associated with vascular inflammation, endothelial dysfunction, platelet activation, and vascular stiffness.⁸

Both cortisol and aldosterone can activate the MR. However, the enzyme 11-HSD2 (11-beta hydroxysteroid dehydrogenase type 2) metabolizes cortisol to the inactive cortisone and thereby protects the MR from activation by cortisol, whereas 11-HSD type 1 metabolizes cortisone to cortisol with the opposite consequence.¹² Elderly hypertensives have a decrease in 11-HSD2 activity in the kidneys and central nervous system (CNS) but an increase in 11-HSD1 activity in the systemic and cerebral vasculature.¹³ These changes in 11-HSD in the elderly should promote activation of the MR by cortisol. Reactive oxygen species accumulate in tissues during aging and can cause inflammation and inhibition of 11-HSD2.¹⁴ Indeed, a study of 1814

normal subjects found a progressive increase in 24-hour urinary free cortisol excretion with age >50 years that could represent the effects of reduced 11-HSD2 or increased 11-HSD1.¹² Postreceptor signaling by the MR is enhanced by the small *p* family GTPase Rac1, whose epigenetic expression is enhanced by reactive oxygen species.^{15,16} Thus, in the elderly, MR signaling in the heart, kidneys, and CNS may be enhanced by a combination of increased MR expression, enhanced MR activation by cortisol and enhanced MR signaling by Rac-1, all of which can be driven by age-related increases in tissue reactive oxygen species and inflammation.

The MR in the brain stem can be activated by aldosterone or cortisol to drive the sympathetic nervous system.¹⁷ The activity of the sympathetic nervous system increases with age and contributes to hypertension, widespread inflammation, cardiovascular disease, and kidney disease. P-glycoprotein acts at the blood-brain barrier (BBB) as an active cell membrane efflux pump for many endogenous compounds, including aldosterone. Its activity declines with age¹⁸ contributing to an enhanced deposition of circulating aldosterone in the brain in older subjects.¹⁹ Moreover, the permeability of the BBB increases in hypertensives because of prolonged small blood vessel disease.²⁰ MRs within the brain contribute to CNS inflammation and oxidative stress, sympathetic nervous system activation, and hypertension, and likely also to the increased risk of stroke and cognitive impairment with age.²¹ Spironolactone preserves BBB integrity and has been suggested to reduce cognitive impairment in CKD.²² Thus, aging may increase the penetration of aldosterone into the brain, where it may increase MR expression and activation. This may summate with increased MR activation in the periphery to drive hypertension, vascular inflammation, oxidative stress, endothelial dysfunction, increased renal sodium absorption and K⁺ loss, hypokalemia, CNS microvascular inflammation and thereby the risk of stroke and cognitive impairment.

PREVENTION AND CORRECTION OF HYPOKALEMIA

Diuretic-induced hypokalemia can be prevented or treated by reducing diuretic dose, increasing dietary K⁺, administration of a K⁺ replacement such as KCl, or by the use of distal diuretics such as amiloride or an MRA. While increased dietary K⁺ reduces blood pressure,²³ it increases SK only modestly, whereas oral KCl increases SK more consistently.² Hypomagnesemia is also provoked by diuretics and prevents correction of hypokalemia. Therefore, hypomagnesemia requires correction. A study in hypokalemic hypertensives taking a thiazide reported that spironolactone or triamterene caused dose-dependent increases in SK, but KCl failed to correct hypokalemia even at a daily dose of >60 mmol

because much of the administered KCl was excreted in the urine.⁵ Moreover, normalization of SK in the elderly with KCl would not prevent MR activation by aldosterone, cortisol, or Rac1. This would require the use of an MRA. Since MRAs may take some weeks to normalize SK, replacement with KCl will be required in patients with an SK <3.5 meq/L, but an MRA alone could be considered in individuals with mild hypokalemia (3.5–4.0 mmol/L). All patients, and especially the elderly receiving both KCl and an MRA, require careful monitoring of serum electrolytes and renal function.

While an MRA can normalize SK and thereby reduce the risks associated with hypokalemia in the elderly and its associated comorbid MR activation, further clinical studies will be necessary to assess its clinical value compared with K⁺ replacement. Aldosterone synthase inhibitors, while effective in increasing SK, reducing BP, and preventing the activation of the MR by aldosterone, would not be expected to reduce age-associated MR activation by cortisol or Rac1. The nonsteroidal MRA finerenone, while effective in reducing adverse cardiovascular and renal outcomes, has not been shown to cross the BBB²⁴ and thus may fail to antagonize the effects of MR activation by cortisol or Rac1 in the CNS in the elderly. The steroidal MRA spironolactone crosses the BBB,²⁵ reduces cardiovascular mortality in elderly individuals with heart failure, and, in observational studies, reduces stroke in patients with hypertension.²⁶ The fear of inducing hyperkalemia has limited the use and persistence of spironolactone, but these concerns should be minimal in patients with hypokalemia. The steroidal MRA eplerenone is more specific for the MR than spironolactone, is associated with a lower incidence of sexual-related side effects such as gynecomastia, and crosses the BBB, although it is less effective in reducing BP than spironolactone.⁵

In conclusion, spironolactone alone, or with short-term administration of KCl replacement therapy, can prevent or correct diuretic-induced hypokalemia and the risks associated with MR sensitization and activation in elderly and very old individuals with hypertension. A combination of spironolactone with a thiazide, thiazide-like, or a loop diuretic that treats hypertension, such as torsemide with careful monitoring of electrolytes and renal function, might be a good choice for elderly hypertensives to prevent the development of hypokalemia. However, this hypothesis should be tested for efficacy, safety, and cost-effectiveness in adequately powered prospective clinical studies in the relatively large number of elderly and very old individuals with hypertension at risk from the consequences of diuretic-induced hypokalemia.

ARTICLE INFORMATION

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